

## Theory Assignment - 01

### Ans : 01

When we open a website, the browser sends a request to the web server. The server process the request and sends a response back.

Browser → HTTP Request → Web Server

Browser ← HTTP Response ← Web Server

Main steps:

1. User enters a website address.
2. Browser sends HTTP request.
3. Server gets and process the request.
4. Server sends response like HTML, CSS or data.
5. Browser shows the webpage.

### Ans : 02

1. HTML – It is used to make the structure of a webpage. It adds headings, paragraphs, images etc.
2. CSS – It is used for styling the webpage like colour, size and layout.
3. JavaScript – It makes the webpage interactive and can do actions when user click something.
4. Bootstrap – It is a framework which gives ready made styles and helps to make responsive websites.
5. jQuery – It is a JavaScript library which makes some JavaScript work easier.
6. React.js – It is a JavaScript library used to make user interfaces and dynamic webpages.

### Ans : 03

Client-side technology runs in the user's browser. Example: HTML, CSS and JavaScript.

Server-side technology runs on the web server. It handles data and other server work. Example: PHP, Node.js.

Client side is mainly for what user see, while server side works in background.

### Ans : 04

Block-level elements normally take the full width and start from a new line. They are used to make sections of a webpage.

Examples are:

<div>

<p>

<h1>

They can also contain other elements inside them.

### Ans : 05

Semantic elements tell what the content is about. They make HTML more clear and easy to understand.

Examples are <header>, <nav>, <main>, <section> and <footer>.

They also help search engines and make the page more organised.

### Ans : 06

```
<table border="1">
  <tr>
    <th>Sr. No.</th>
    <th>Name</th>
    <th>Designation</th>
    <th>Department</th>
  </tr>
  <tr>
    <td>1</td>
    <td>John</td>
    <td>Manager</td>
    <td>Sales</td>
  </tr>
  <tr>
    <td>2</td>
    <td>Smith</td>
    <td>Sr. Manager</td>
    <td>Marketing</td>
  </tr>
  <tr>
    <td>3</td>
    <td>Jane</td>
    <td>Manager</td>
    <td>HR</td>
  </tr>
</table>
```

### Ans : 07

```
<ul>
  <li>River
    <ul>
      <li>Ganga</li>
      <li>Yamuna</li>
      <li>Brahmaputra</li>
      <li>Narmada</li>
    </ul>
  </li>
</ul>
```

### Ans : 08

There are three types of CSS:

1. Inline CSS – CSS is written inside the HTML tag.  
Example: `<p style="color:red;">Hello</p>`
2. Internal CSS – CSS is written inside the `<style>` tag in the HTML file.
3. External CSS – CSS is written in a separate .css file and linked with HTML. This is useful for bigger websites.

### Ans : 09

HTML makes the basic structure of a webpage.

CSS makes the webpage look good. It controls colours, fonts and layout.

JavaScript adds actions and interaction to the webpage. So, HTML is structure, CSS is design and JavaScript is behaviour.

### Ans : 10

- a. Bootstrap gives ready made CSS classes and components. It helps make websites faster and responsive.
- b. jQuery is a JavaScript library. It was widely used because it made DOM changes, events and other JavaScript tasks easier.
- c. React.js is a JavaScript library for making user interfaces. It helps build dynamic pages using reusable components.